

TOP060SOFT10802160-J0

5.99" FHD

Product Specification

Rev. P0



Customer Name

:

Product Name

: 5.99" FHD1080 AMOLED

Model Name

: CD-2

Description

: 5.99" FHD (1080×2160) 16M Color

Proposed by			Customer's Approval
Designed	Checked	Approved	

Shenzhen Top Electronic Parts Co., Limited

Revision History

Rev.	ECN No.	Description of Change	Date	Prepared
P0	-	-.Initial issue	2017.07.12	

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1. General Description

1-1. Introduction

BOE 5.99" FHD is a color active matrix AMOLED module using Low Temperature Poly-silicon TFT's (Thin Film Transistors) as active switching devices. This module has a 5.99inch diagonally measured active area with FHD resolutions (1080horizontal by 2160vertical pixel arrays). Each pixel is divided into RED and GREEN dots, or BLUE and GREEN dots, and two pixels share RED or BLUE dots which are arranged in vertical stripe and this module can display 16.7M colors.

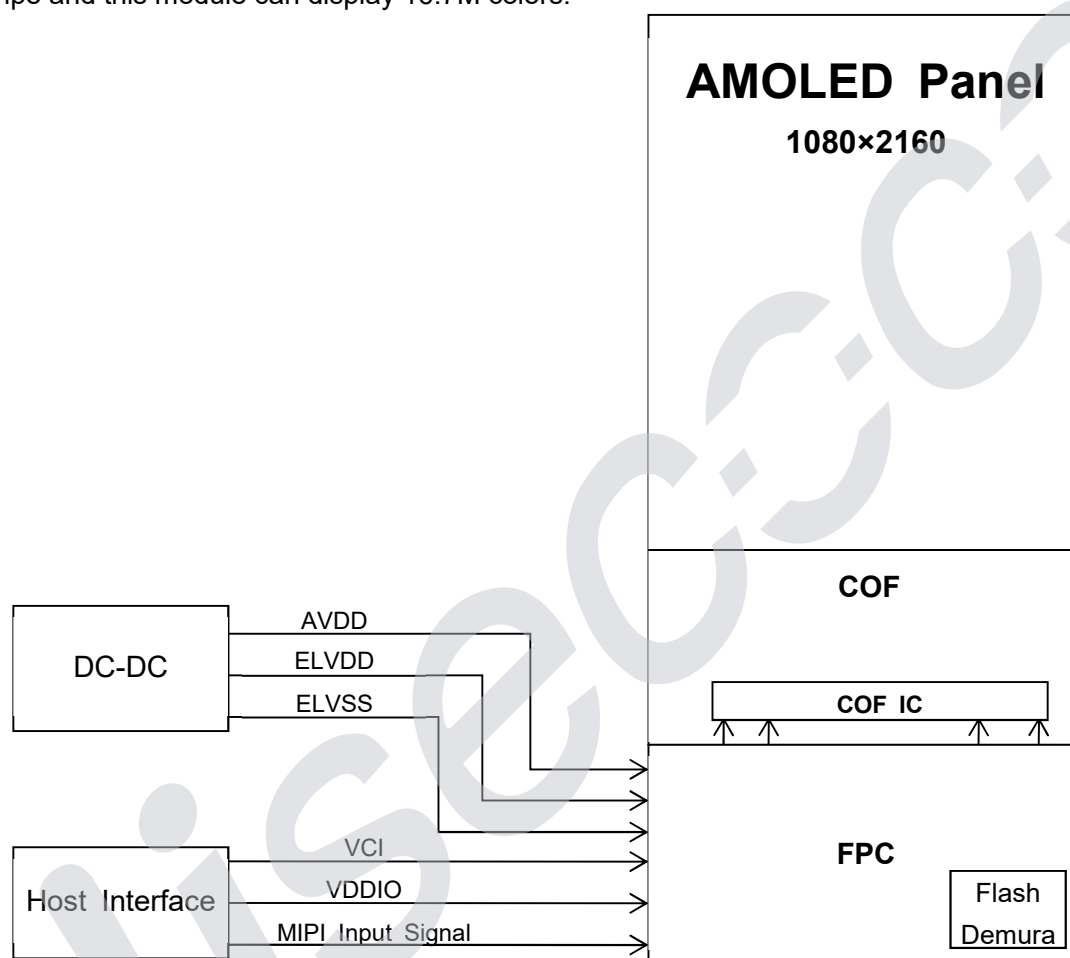


Figure 1

1-2. Features

- 1) AMOLED Display type with 3D Cover Glass
- 2) Display Format : 5.99" FHD GGRB : 1080×2160
- 3) Interface : MIPI-DSI 4 lanes
- 4) Driver IC : SW43404
- 5) Polarizer : CPOL(with PSA)+Flexible Touch Sensor

1-3. Application

- Smart Mobile Phone

2. Specification

2-1. Panel Specification

Table 1

Item	Specifications	Unit	Remark
Panel outline	70.04×139.53×0.13	mm	
Number of dots	1080×2160		18:9
Active area	68.04×136.08	mm	
Diagonal Inch	5.99	inch	
Pixel pitch	63	um	
Pixel Arrangement	GGRB		

2-2. Mechanical Specification

Table 2

Item	Specifications	Unit	Remark
MDL outline	71.35×150.42	mm	
View area	68.63×137.08	mm	
Weight	TBD		
Total Thickness	1.416	mm	

3. Absolute Maximum Ratings

Table 3

Item	Symbol	Min.	Max.	Unit	Note
I/O Voltage	VDDIO	1.62	1.98	V	
Operation Voltage	VCI	2.60	3.60	V	
EL Driving Voltage	ELVDD	4.40	4.80	V	
	ELVSS	-5.00	-1.40	V	
Supply voltage (TSP)	VCC	2.7	3.6	V	
Operating temperature	Topr	-20	70	°C	
Storage temperature	Tstg	-40	80	°C	

4. Electrical Characteristics

4-1 Power Consumption of Display Panel and Touch panel



Test Condition: Temp=25±2°C

Table 4

Item		Symbol		Condition	Min.	Typ.	Max.	Unit	Remark
ELVDD		ELVDD		-	4.40	4.60	4.80	V	
ELVSS		ELVSS		-	-5.0	-2.40	-1.40	V	Controlled by DDIC
AVDD		AVDD		-	5.40	6.80	7.70	V	
VCI		VCI		-	2.60	3.00	3.60	V	
VDDIO		VDDIO		-	1.62	1.80	1.98	V	
Current Consumption (Display)	Display on mode	IC	VCI	Frame F=60Hz White pattern	1.5	2.0	2.5	mA	Full White 390 nit
			VDDIO		20.0	30.0	40.0	mA	
			AVDD		10	20	30	mA	
	Sleep in mode	Panel	EL	Sleep mode	180	220	260	mA	
		IC	VCI		-	0.1	0.3	mA	
			VDDIO		1.0	1.5	2.0	mA	
Frame Frequency		F _{frm}		-	59	60	61	Hz	
Vin		Vin(DC DC Input)		White Mode L255	-	-	-	mA	Vin=4v
					-	-	-	mA	Vin=3.7v

Notes :

1. The value is just the reference value. The customer may optimize the setting value.
2. The current of Vin is just the reference value, because it depends on the efficiency of Power IC.

4-2 Power Consumption of Display Panel and Touch panel

Power Supply: TSP_3.3V =3.3V, TSP_1.8V =1.8V, VCI_3.0V =3.0V, VDD_1.8V =1.8V, VLIN1=6.8V

Report Rate: 80Hz @ Display panel Frame Frequency = 60Hz

Table 5

Work Mode	Item	Symbol	Value		Unit	Remark
			Typ	Max		
Sleep Mode	Current of TSP_1.8V	ITSP_1.8V	10	-	uA	
	Current of TSP_3.3V	ITSP_3.3V	50	-	uA	
Idle Mode	Current of TSP_1.8V	ITSP_1.8V	500	-	uA	
	Current of TSP_3.3V	ITSP_3.3V	10	-	mA	
No finger	Current of TSP_1.8V	ITSP_1.8V	1.95	-	mA	
	Current of TSP_3.3V	ITSP_3.3V	20	-	mA	
1 Finger	Current of TSP_1.8V	ITSP_1.8V	7.65	-	mA	
	Current of TSP_3.3V	ITSP_3.3V	22	-	mA	
5 Finger	Current of TSP_1.8V	ITSP_1.8V	10.05	-	mA	
	Current of TSP_3.3V	ITSP_3.3V	23	-	mA	
10 Finger	Current of TSP_1.8V	ITSP_1.8V	11.85	-	mA	
	Current of TSP_3.3V	ITSP_3.3V	25	-	mA	

5. Electro-optical Characteristics

Table 6

Item		Symb ol	Temp	Condition	Min.	Typ.	Max.	Unit	Note
Brightness			25°C	Normal (W L255)	351	390	429	Cd/m²	Center Brightness
Uniformity			25°C		75	85	-	%	(1)
Contrast ratio		K	25°C	Φ=0°,θ=0°	-	100000:1	-	-	(2) Based on CA-310 (Probe 10 Φ)
Color of CIE coordinate	White	x	25°C	Φ=0°,θ=0° CIE1931	0.2686	0.2988	0.3305	-	Color of CIE coordinate
		y			0.2978	0.3158	0.3347	-	
	Red	x			0.635	0.665	0.695	-	
		y			0.304	0.334	0.364	-	
	Green	x			0.207	0.237	0.267	-	
		y			0.681	0.711	0.741	-	
	Blue	x			0.115	0.135	0.155	-	
		y			0.032	0.0520	0.072	-	
Color Gamut			25°C	NTSC , CIE1931	86	101	116	%	(3)
White color shift			25°C	Φ=30°,θ=30°	-	≤5.2	-	JNCD	
Viewing Angle			25°C	Up/Down/Right/Left CR ratio ≥10	75	80	-	deg	(4)
Cross Talk			25°C	Background: gray128	-	-	2%		(5)
Gamma			25°C	-	2.0	2.2	2.4		
Life time			25°C	Light on for 240 hrs.	luminance decrease≤6%			cd/m²	(6)

Notes :

(1) Uniformity Measuring Point: Typical luminance: based on different customers different requirements, for example 350 cd/m², 380 cd/m², 390 cd/m² etc.

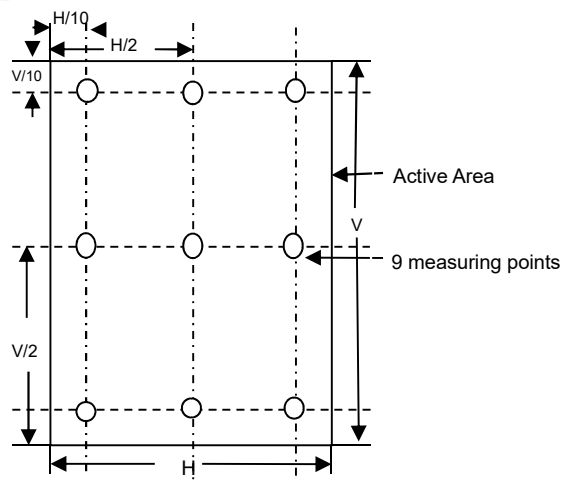


Figure 2. Uniformity Measuring Point

$$\text{Uniformity} = L_{\min} / L_{\max} \times 100\%$$

(2) Definition of contrast ratio(K)

$$CR = \frac{\text{Luminance When Display Panel is at "White" State}}{\text{Luminance When Display Panel is at "Black" State}}$$

(3) Definition of Color of CIE Coordinate and NTSC Ratio

The test condition is at 25°C and measured on the surface of Display panel module.

- Measurement equipment: CS2000 or similar equipment.
- The Color Coordinate (CIE 1931/CIE 1976) measure the center of active area of the module.

$$S = \frac{\text{Area of RGB Triangle}}{\text{Area of NTSC Triangle}} \times 100\%$$

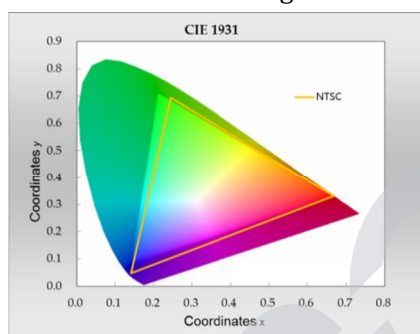


Figure 4. Color Coordinate

(4) Viewing Angle measuring system

Refer to the graph below marked by θ and ϕ

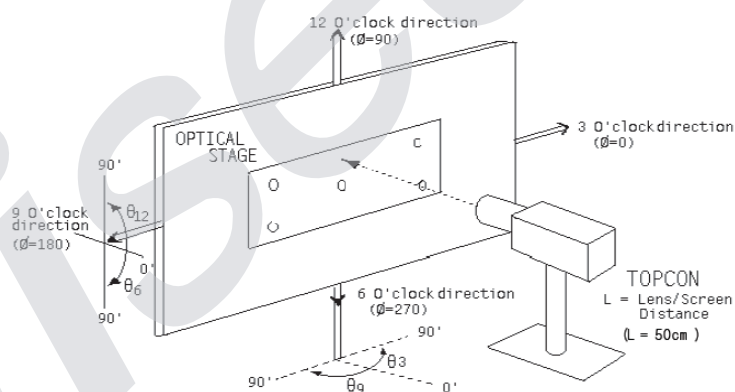


Figure 5. Viewing Angle measuring system

(5) Crosstalk measurement shall be done at the center of the different pattern and the result shall be calculated as follow formula.

- measure luminance at the center.
- calculate cross talk as below equation:

$$\text{Crosstalk}(V = \max) = \frac{\text{Luminance at center of V}}{\text{Luminance at center of H}} \times 100, \quad \text{Crosstalk}(H = \max) = \frac{\text{Luminance at center of H}}{\text{Luminance at center of V}} \times 100$$

$$\text{Crosstalk}(H = \max) = \frac{\text{Luminance at center of H}}{\text{Luminance at center of V}} \times 100, \quad \text{Crosstalk}(V = \max) = \frac{\text{Luminance at center of V}}{\text{Luminance at center of H}} \times 100$$

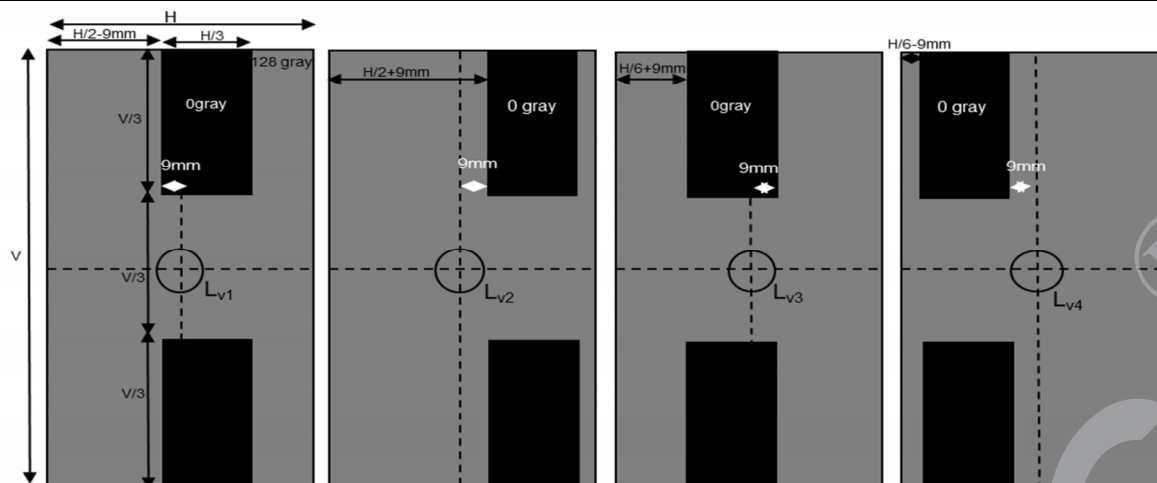


Figure 6. Vertical crosstalk measuring pattern

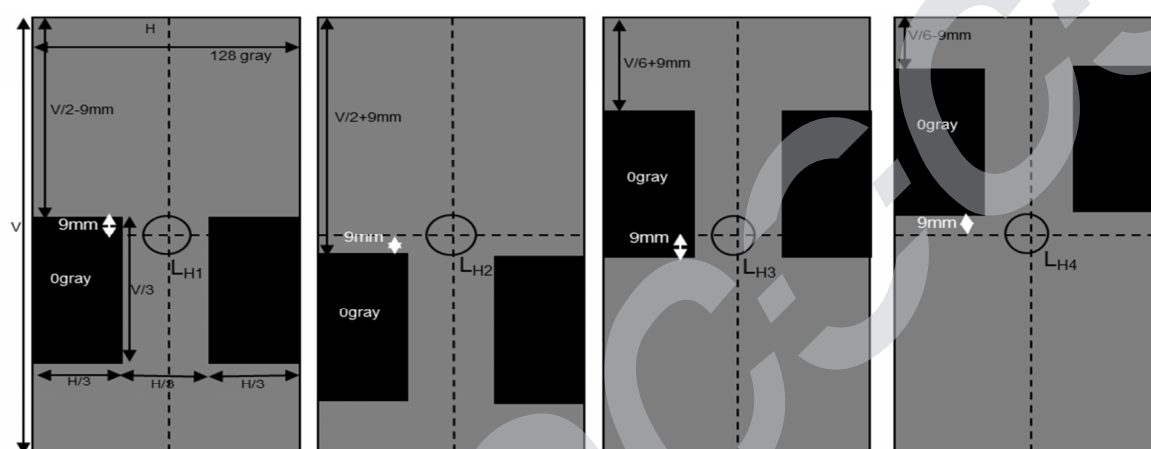


Figure 7. Horizontal crosstalk measuring pattern

(6) OLED life time

Test samples 8pcs;

At room temperature(25℃), light the module with typical value brightness, display a white pattern.

Keep working 240 hours, luminance decrease ≤ 6%.

The test data must pass the specification.

6. FPC Pin Assignment

Main FPC assignment- AMOLED Panel Input/output Signal Interface.

Recommended connector: 24-5857-048

According to the customer demand

Table 7

No.	Name	No.	Name
1	TSP-INT	25	SWIRE
2	TSP-SCL	26	AVDD_EN
3	TSP-SDA	27	TE
4	GND	28	ERR_FG
5	D2P	29	RESX
6	D2N	30	OPEN
7	GND	31	OPEN
8	DIP	32	OPEN
9	DIN	33	OPEN
10	GND	34	OPEN
11	CLKP	35	GND
12	CLKN	36	OPEN
13	GND	37	ELVSS
14	D0P	38	ELVSS
15	D0N	39	ELVSS
16	GND	40	OPEN
17	D3P	41	ELVDD
18	D3N	42	ELVDD
19	GND	43	ELVDD
20	VDDIO_1.8V	44	OPEN
21	OPEN	45	GND
22	VCI_3.0V	46	TSP_AVDD_3.3V
23	OPEN	47	TSP_RESET
24	AVDD_6.8V	48	TSP_DVDD_1.8V

<Pin layout of B-to-B contact pads>

7. AC Characteristics (For reference only)

7-1. MIPI DSI Characteristics

7-1-1. DC Characteristics for MIPI DSI

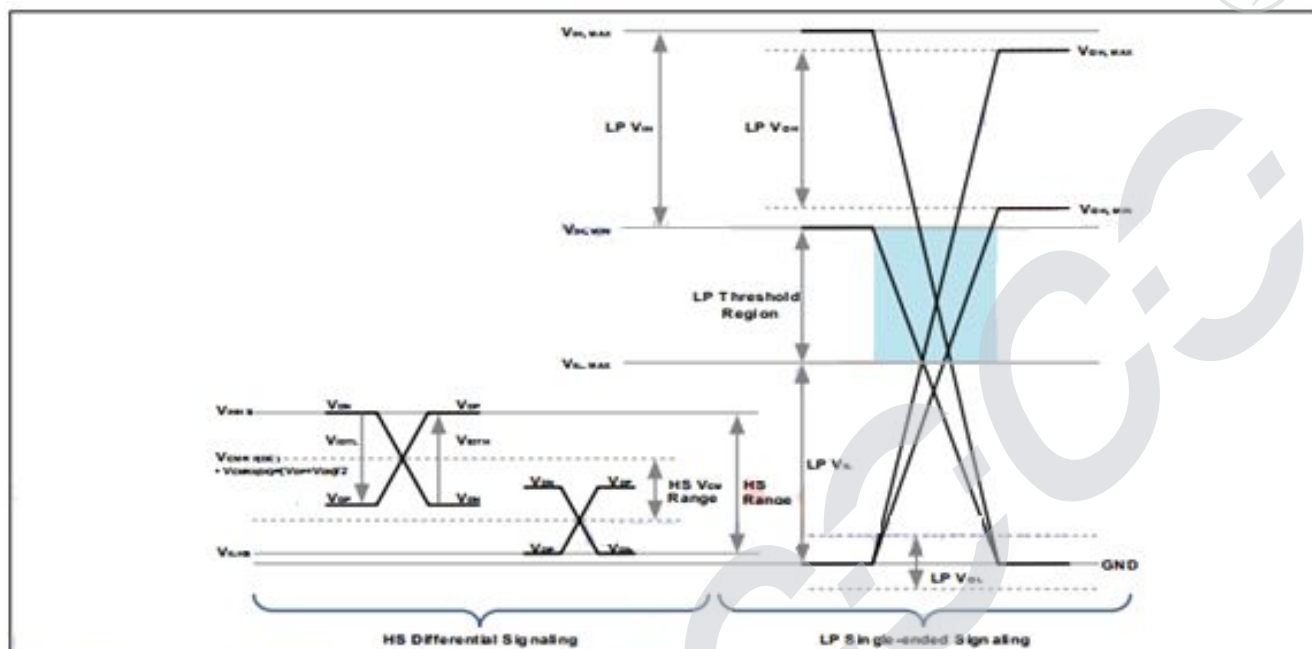


Figure 8 MIPI DSI Signaling Levels

Table 8

Signal	Symbol	Parameter	Min.	Typ.	Max.	Unit
HS_RX	V_{IDTH}	Differential input high threshold	-	-	70	mV
	V_{IDTL}	Differential input low threshold	-70	-	-	
	V_{IHHS}	Single-ended input high voltage	-	-	460	
	V_{ILHS}	Single-ended input low voltage	-40	-	-	
	$V_{TERM-EN}$	Single-ended threshold for HS termination enable	-	-	450	
	$V_{CMRX(DC)}$	Common-mode voltage HS receive mode	70	-	330	
	Z_{ID}	Differential input impedance	-	100	-	Ω
LP_RX	V_{IL}	Logic0 voltage not in ULP State	-	-	550	
	V_{IH}	Logic1 input voltage	880	-	-	mV
	V_{LEAK}	I/O leakage current	-10	-	10	μA
LP_TX	V_{OL}	The venin output low level	-50	-	50	mV
	V_{OH}	The venin output high level	1.1	1.2	1.3	V
	Z_{OLP}	Output impedance of LP transmitter	110	-	-	Ω

7-1-2. MIPI DSI High-Speed RX Clock and Data-Clock Timing

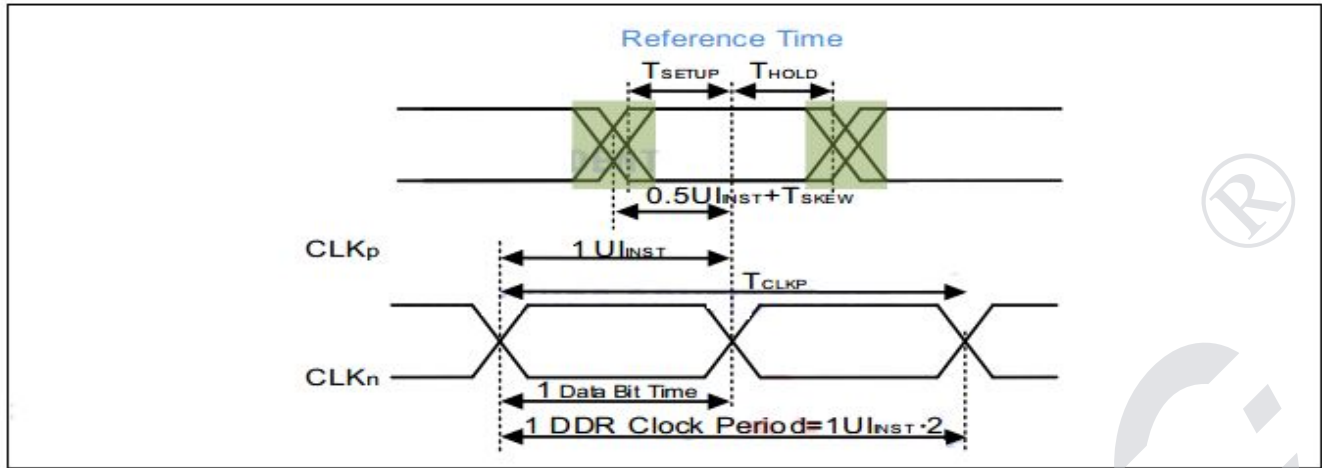


Figure 9 MIPI Data to Clock Timing Definitions

Table 9

Symbol	Parameter	Condition	Min.	Typ.	Max.	Unit
F_{DSICLK}	DISCLK Frequency	VDDI=1.6 5~3.3V	T.B.D.		500	MHz
T_{CLKP}	DSICLK Cycle time		1		10	Ns
T_{DSIR}	DSI Data Transfer Rate		T.B.D.		1000	Mbps
$T_{SKEW[TX]}$	Data to Clock Skew		-0.15		0.15	UI_{INST}
T_{SETUP}	Data to Clock Setup time		0.15	-	-	UI_{INST}
			0.15	-	-	ns
T_{HOLD}	Data to Clock Hold time		0.15	-	-	UI_{INST}
			0.15	-	-	ns
UI_{INST}	UI instantaneous		1		12.5	ns

7-1-3. Global Operation Timings

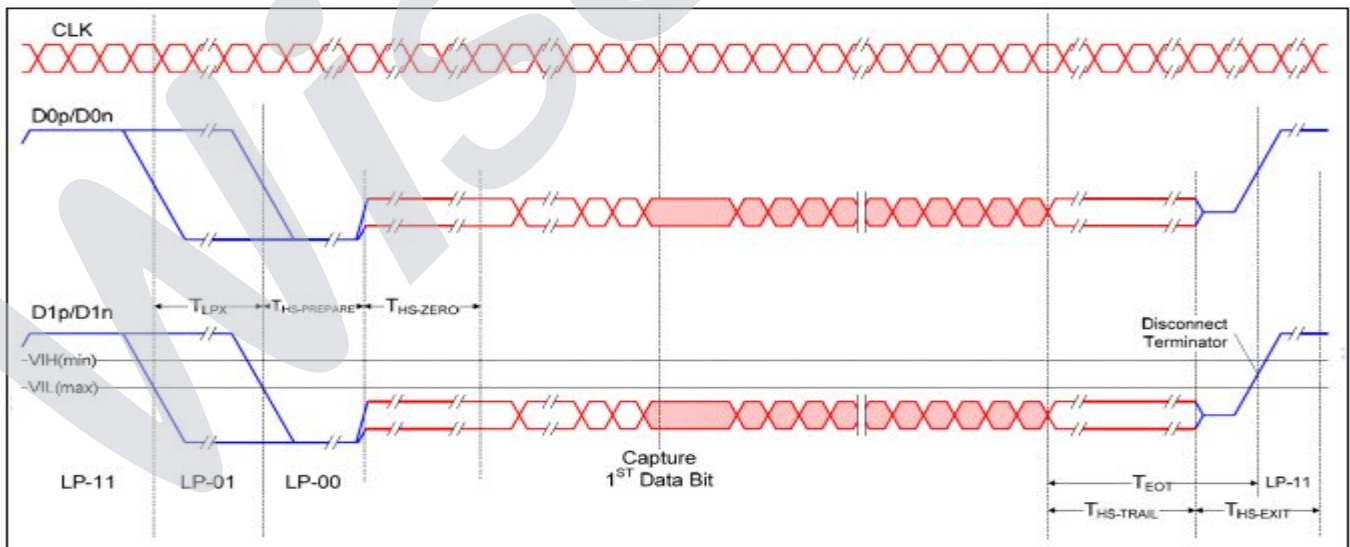


Figure 10 MIPI HS Data Transmission in Bursts

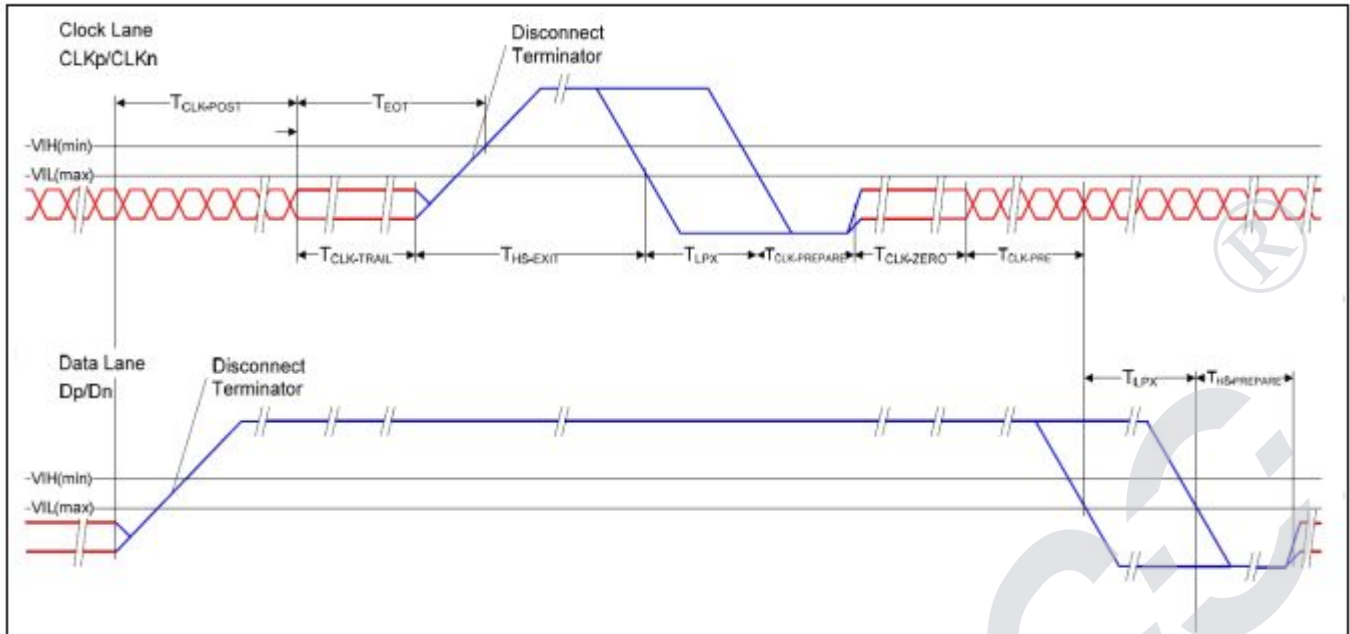


Figure 11 MIPI switching the Clock Lane between Clock Transmission and LP mode

Table 10

Parameter	Description	Spec.			Unit
		Min.	Typ.	Max.	
$T_{CLK-MISS}$	Timeout for receiver to detect absence of Clock transitions and disable the Clock Lane HS-RX.			60	ns
$T_{CLK-POST}$	Time that the transmitter continues to send HS clock after the last associated Data Lane has transitioned to LP Mode. Interval is defined as the period from the end of $T_{HS-TRAIL}$ to beginning of $T_{CLK-TRAIL}$	$60\text{ ns} + 52 \cdot UI$	-	-	ns
$T_{CLK-PRE}$	Time that the HS clock shall be driver by the transmitter prior to any associated Data Lane beginning the transition from LP to HS mode.	8	-	-	UI
$T_{CLK-PREPARE}$	Time that the transmitter drives the Clock Lane LP-00 Line state immediately before the HS-00 Line state starting the HS transmission.	38	-	95	ns
$T_{CLK-SETTLE}$	Time interval during which the HS receiver shall ignore any clock Lane HS transitions, starting from the beginning of $T_{CLK-PREPARE}$.	95	-	300	ns
$T_{CLK-TERM-EN}$	Time for the Clock Lane receiver to enable the HS line termination, starting from the time point when Dn crosses V_{ILMAX} .	Time for Dn to reach $V_{TERM-EN}$	-	38	ns
$T_{CLK-TRAIL}$	Time that the transmitter drives the HS-00 state after the last payload clock bit of a HS HS transmission burst.	60	-	-	ns
$T_{CLK-PREPARE} + T_{CLK-ZERO}$	$T_{CLK-PREPARE} +$ Time to that the transmitter drives the HS-00 state prior to starting the clock.	300	-	-	ns

$T_{D-TERM-EN}$	Time for the Data Lane receiver to enable the HS line termination, starting from the time point when Dn crosses V_{ILMAX} .	Time for Dn to reach $V_{TERM-EN}$	-	35ns+4*UI	-
T_{EOT}	Transmitted time interval from the start of $T_{HS-TRAIL}$ or $T_{CLK-TRAL}$, to the start of the LP-11 state following a HS burst.	-	-	105 ns + n*12*UI	-
$T_{HS-EXIT}$	Time that the transmitter drives LP-11 following HS burst.	100	-	-	ns
$T_{HS-PREPARE}$	Time that the transmitter drives the Data Lane LP-00 Line state immediately before the HS-00 Line state starting the HS transmission.	40ns+4*UI	-	85ns+6*UI	ns
$T_{HS-PREPARE} + T_{HS-ZERO}$	$T_{HS-PREPARE}$ + Time that the transmitter drives the HS-0 state prior to transmitting the Sync sequence.	145 ns + 10*UI	-	-	ns
$T_{HS-SETTLE}$	Time interval during which the HS receiver shall ignore any Data Lane HS transitions , starting from the beginning of $T_{HSPREPARE}$	85 ns + 6*UI	-	145 ns + 10*UI	ns
$T_{HS-SKIP}$	Time interval during which the HS-RX should ignore any transitions on the Data Lane, following a HS burst. The end point of the interval is defined as the beginning of the LP-11 state following the HS burst.	40	-	55 ns + 4*UI	ns
$T_{HS-TRAIL}$	Time that the transmitter drives the flipped differential state after last payload data bit of a HS transmission burst	Max(n*8*UI, 60ns + n 4* UI)	-	-	ns
T_{INIT}	-	-	-	-	-
T_{LPX}	Transmitted length of any Low-Power state period	-	56.6	-	ns
Ratio T_{LPX}	Ratio of $T_{LPX(MASTER)}/T_{LPS(SLAVE)}$ between Master and Slave side	2/3	-	3/2	
T_{TA-GET}	Time that the new transmitter drives the bridge state (LP-00) after accepting control during a Link Turnaround.	5*T _{LPX}			ns
T_{TA-GO}	Time that the new transmitter drives the bridge state (LP-00) before releasing control during a Link Turnaround.	4*T _{LPX}			ns
$T_{TA-SURE}$	Time that the new transmitter waits after the LP-10 state before transmitting the Bridge state (LP-00) during a Link turnaround.	T_{LPX}	-	2*T _{LPX}	ns
T_{WAKEUP}	Time that a transmitter drives a Mark-1 state prior to a Stop state in order to initiate an exit from ULPS.	1			ms

7-1-4. AC Characteristics of MIPI DSI Characteristics

Table 11

Symbol	Parameter	Min.	Typ.	Max.	Unit
Thost-enable	Host output enable time	0	-	24*t-bit	ns
Thost-disable	Host output disable time, entire length of the Turnaround 1 field	0	-	24*t-bit	
Tclient-enable	Client output enable time, entire length of the Turnaround 1 field	0	-	24*t-bit	
Tclient-disable	Client output disable time, measured from the end of the last bit of the Turnaround 2 field.	0	-	24*t-bit	

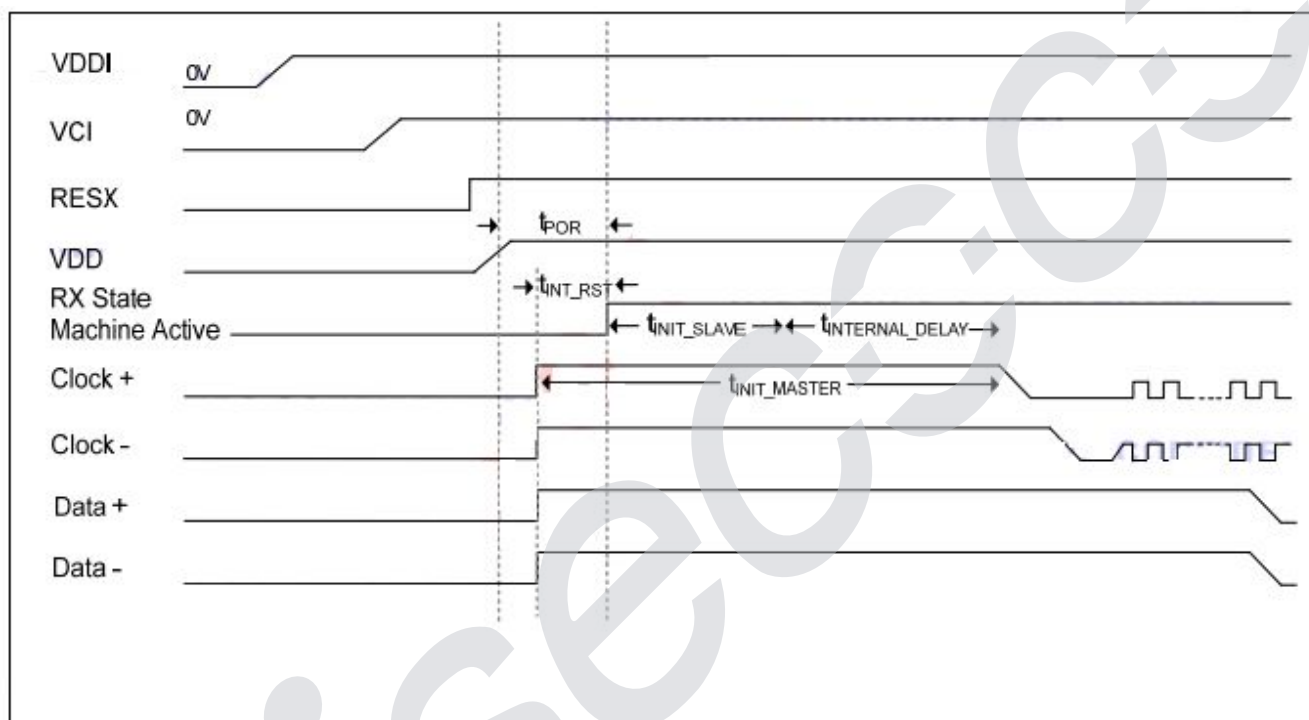


Figure 12 MIPI lanes status when reset operation

7-2. Touch Panel I2C Timing Characteristics

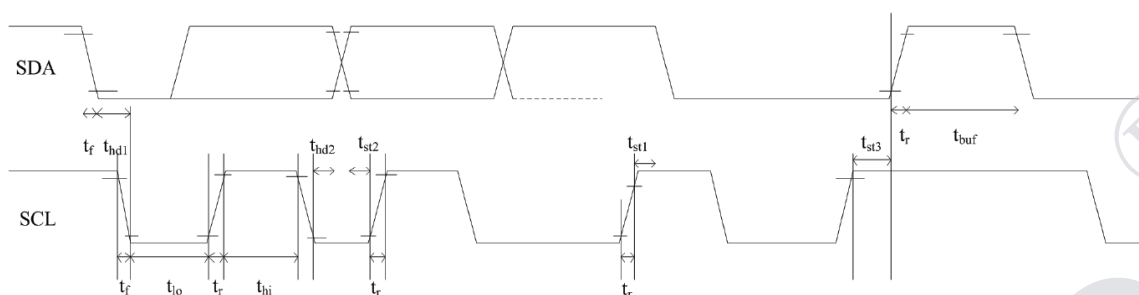


Figure 13

Table 12

Parameter	Symbol	Min.	Max.	Unit
SCL Low Period	t_{lo}	1.3	-	us
SCL High Period	t_{hi}	0.6	-	us
SCL setup time for START condition	t_{st1}	0.6	-	us
SCL setup time for STOP condition	t_{st3}	0.6	-	us
SCL hold time for START condition	t_{hd1}	0.6	-	us
SDA setup time	t_{st2}	0.1	-	us
SDA hold time	t_{hd2}	0	-	us

7-3. Touch panel external reset timing

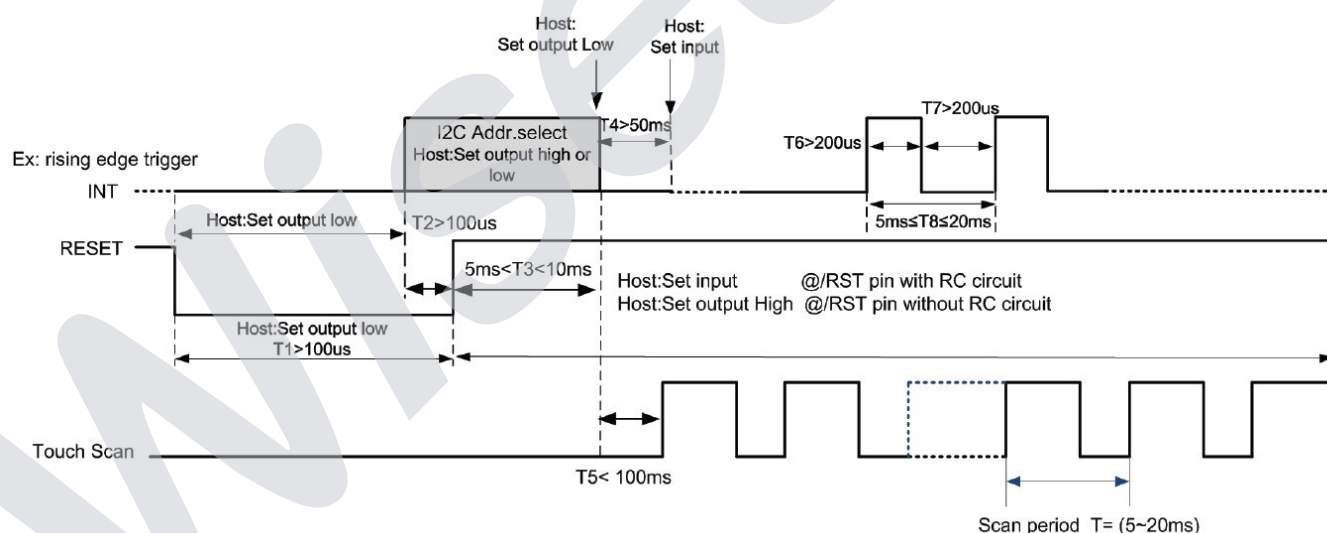


Figure 14

7-4 Touch Specification

Table 13

No.	ITEM		SPEC	REMARK
1	Touch IC			TBD
2	Communication Protocol to Host		I2C	
3	Multi-Finger		10	
4	I2C Address			TBD
5	Touch Origination Dot		Left up	
6	Performance	Accuracy	Edge area<1.5mm Center area<1.0mm	Note (1)
		Linearity	Edge area <1.5mm Center area <1.0mm	Note (2)
		Point Sensitivity	≤Φ4mm	Note (3)
		Move Sensitivity	≤Φ4mm	Note (4)
		Jitter	Edge area <0.30mm Center area <0.20mm	Note (5)
		Water Proof	Support	Note (6)
7	Structure		GF	Touch on Film
8	Sensor Pitch(A)		4.563(H)*4.495(V)	
9	Connector type /No.		FOF	
10	Low Temperature		-10℃	Handset can response exactly (No ghost finger, No missing finger, etc)
11	Obvious ITO etching pattern		Invisible	
12	power consumption		Active model: 82mW	
			Standby mode: 33mW	
			Sleep model: 165uW	

Note (1): Accuracy is determined by a comparison of the actual copper position and the reported position when the copper touch on the surface of touch.

1. Test Condition : Handset is on the insulated table.

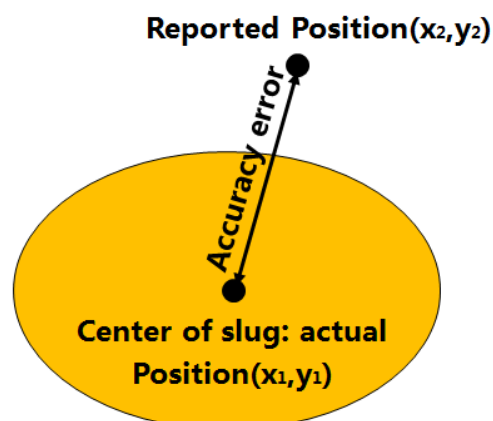
2. Measurement equipment: Arm of robot with 7mm diameter copper.

3. Test procedure: Test touch panel with 7*9 points , each point 10 time.

4. The Accuracy is calculated by using following formula:

Calculate every distance from reported position to actual position (Each point contains 10 reported position dates)

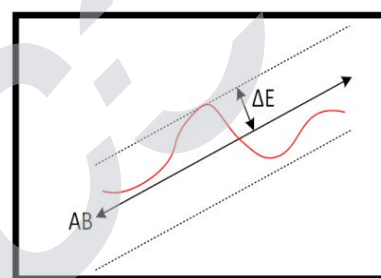
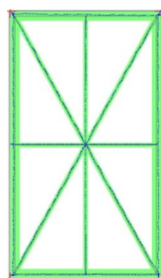
Accuracy Error = square root $[(x_i - x_0)^2 + (y_i - y_0)^2]$ (i=1,2...10)



Note (2): Linearity is defined as the difference between reported finger positions versus the least square fitted line as the finger moves linearly across a specified trajectory of the Display panel area.

1. Test Condition : Handset is on the insulated table.
2. Measurement equipment: Arm of robot
3. Test procedure: Draw 8 line with 30mm/s by 7mm copper.
4. The Precision is calculated by using following formula:

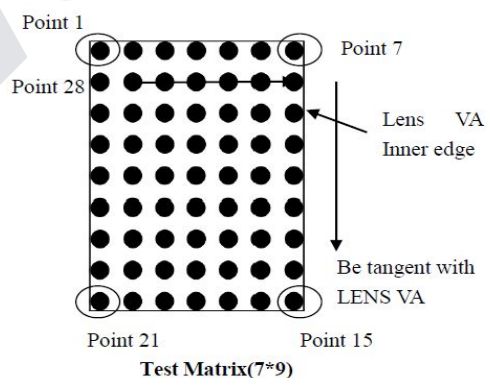
Calculate the max ΔE for each line



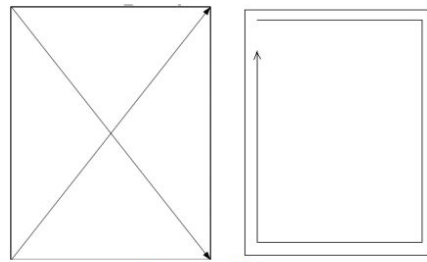
Note (3): Point sensitivity is determined by the minimum size finger that touch panel can detect. When the minimum size finger touch on the surface of touch, the touch can report to host exactly.

1. Test Condition : Handset is on the insulated table.
2. Measurement equipment: Arm of robot with 6mm diameter copper.
3. Test procedure: Test touch panel with 7*9 points , each point 10 times
4. The point sensitivity is calculated by using following formula:

Report Rate = Reported points/280*100%



Note (4): Move sensitivity is determined by the minimum size finger that touch panel can detect. When the minimum size finger draw on the surface of touch, the touch can report to host exactly.

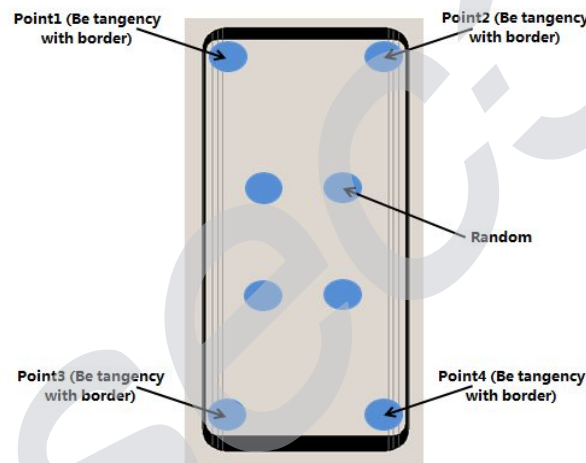


Move sensitivity test

1. Test Condition : Handset is on the insulated table.
2. Measurement equipment: Arm of robot
3. Test procedure: Draw 8 line with 30mm/s by 6mm copper

Standard: No missing point.

Note (5): Jitter is defined as the deltas of reported positions when a conductive copper is in stationary contact with the sensor cover lens. A total of hundred sequential samples are collected with each stationary contact of the Copper with the sensor cover lens.



1. Test Condition : Handset is on the insulated table.
2. Measurement equipment: Arm of robot with 6mm diameter copper.
3. Test procedure: Test 8 points in the touch for 1s
4. The Precision is calculated by using following formula:
Then we will get the result like below (Take point 1 as the example)
 - (1). calculate distance from each reported position to the rest of points
Distance Error = square root $[(x_i - x_j)^2 + (y_i - y_j)^2]$ ($i=1,2,...,120, j=1,2,...,120$)
 - (2). Select the maximum distance error from one to each one
jitter 1= max(error 1,error 2....error 120)
 - (3). Repeat 1to 2 for the other 7 point as the jitter value
 - (4). Select the maximum value as our test result
jitter= max (jitter1 ,jitter 2..... jitter 8)

Note (6): Anti-water:

The presence of moisture on the surface of touch can affect touch performance. Performance will vary based on the amount of moisture and its properties. In the test we will define the basic

requirement in the document.

Method:

Drop: Size : $\varnothing 10$ mm diameter drop, 4 drops

Spray size: 3ml once

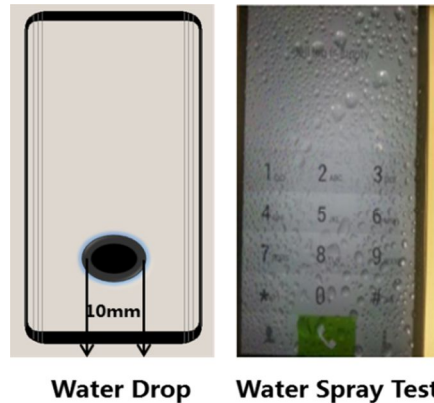
Procedure:

Test 1-drop test:

Step 1: Make 4 drops water on the surface of touch, each drop with 10mm diameter.

Step 2: Test the area (without water area) handwork, and test it again after wiped off water

Step 3: Observe whether the water area report ghost finger



8. Recommended Operating Sequence

8-1. Display Power on/off Sequence

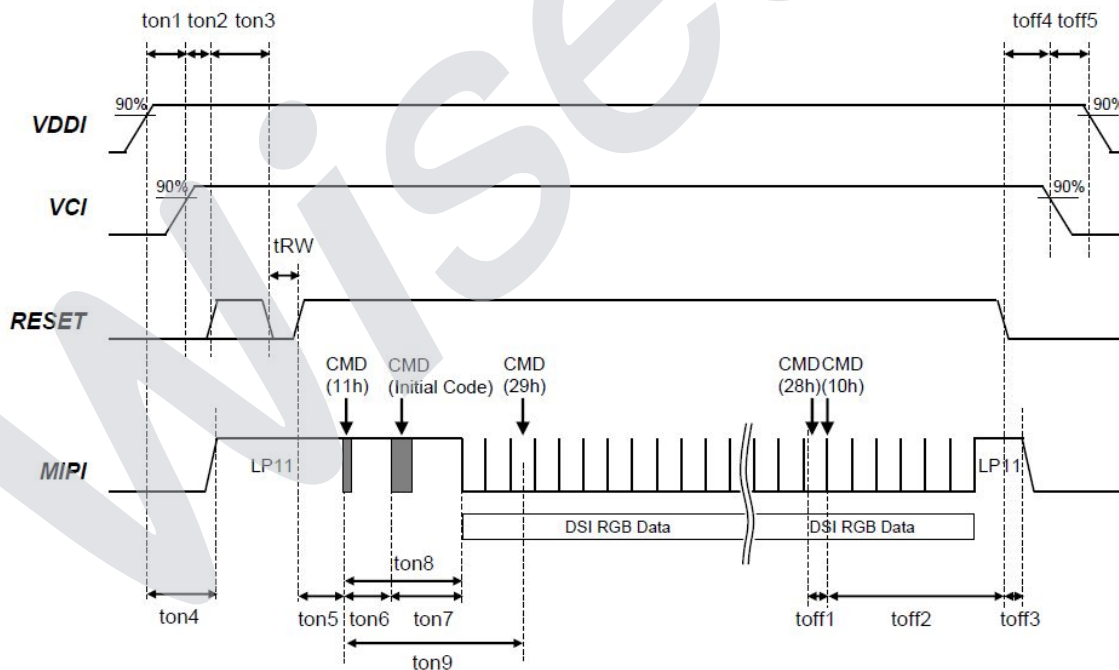


Figure 15 Power On/Off Sequence

Parameter	min	typ	max	unit
ton1	0	-	-	ms
ton2	1	-	-	ms
ton3	9	-	-	ms
ton4	1	-	ton1 + ton2	ms
ton5	20	-	-	ms
ton6	10	-	-	ms
ton7	20	-	-	ms
ton8	30	-	100	ms
ton9	0	-	-	ms
tRW	1	-	-	ms
toff1	0	-	-	ms
toff2	150	-	-	ms
toff3	0	-	-	ms
toff4	0	-	-	ms
toff5	0	-	-	ms

Figure 16

8-2. Touch Panel Power on Sequence

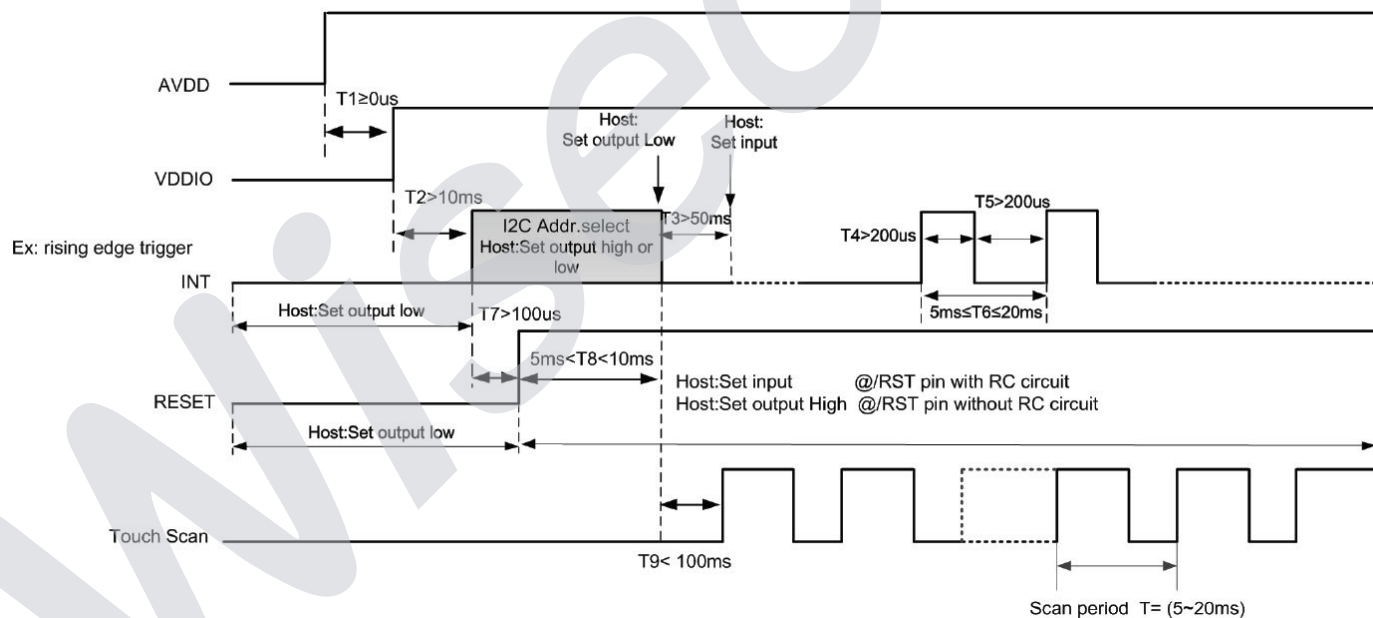


Figure 17

9. Outline Information

9-2-1.Main FPC Schematic Diagram

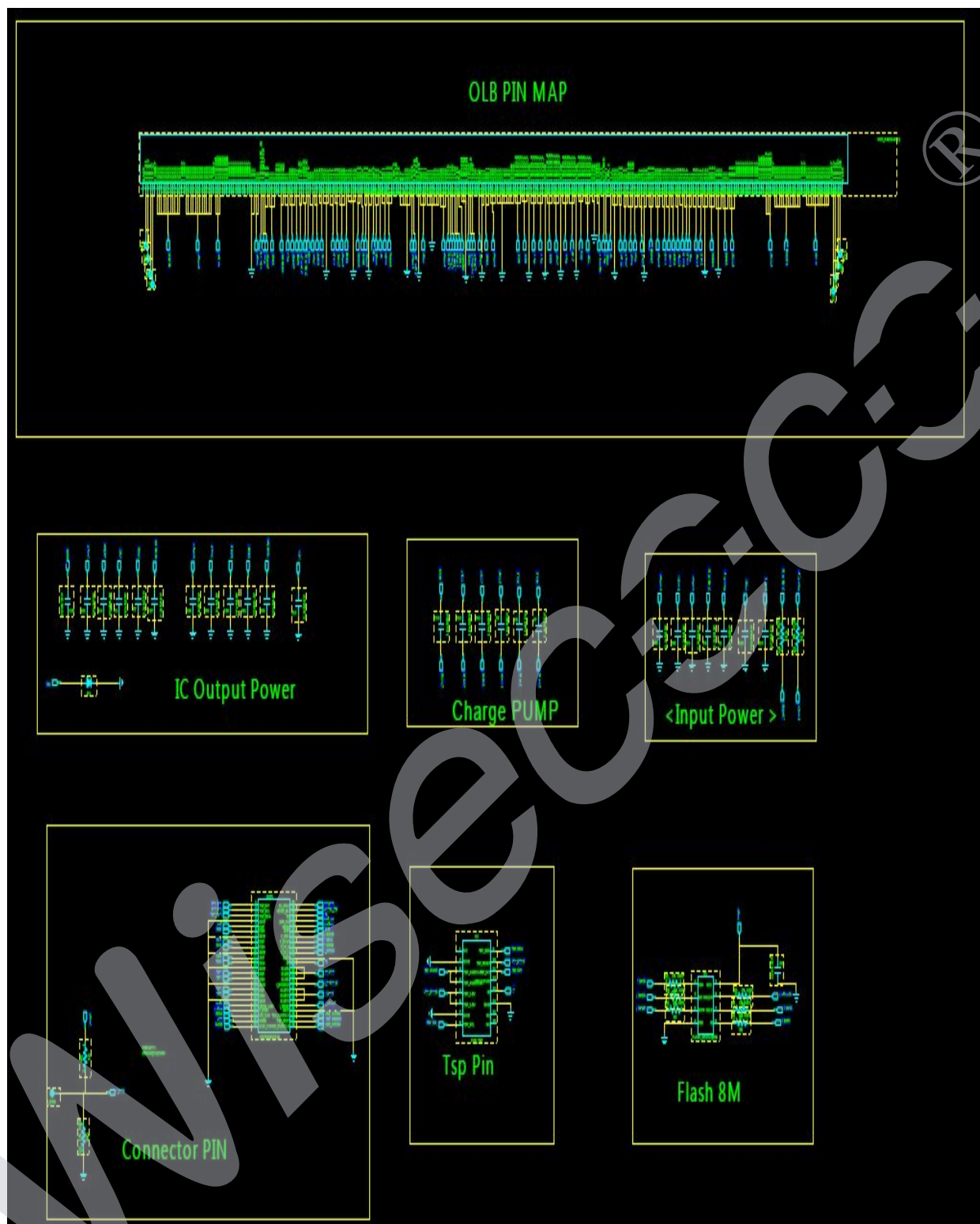


Figure 19

9-2-2.Main FPC Electronic Part List

Table 14

TABLE 1									
CATEGORY	REFERENCE	SPECIFICATION	VALUE	耐压	封装	材质	Q'ty	MAKE	
SPEC. NUMBERS	TOP060SOFT10802160-J0							PAGE 24/ 32	

		Shenzhen Top Electronic Parts Co., Limited				REV.	ISSUE DATA	
						P0	2017.08.03	
				值				R
CAPACITOR	C7		1uF	6.3V	0201	X5R	1	
	C4,C14,C22,C27		2.2uF	6.3V	0201	X5R	4	
	C31		22nF	6.3V	0201	X5R	1	
	C15		4.7uF	6.3V	0201	X5R	1	
	C30		1.0uF	10V	0402	X5R	1	
	C8-9,C11-13,C29		1.0uF	16V	0402	X5R	6	
	C35,C38,C40		2.2uF	16V	0402	X5R	3	
	C2,C5,C20-21,C25		2.2uF	25V	0402	X5R	5	
	C1,C6,C10, C26		4.7uF	16V	0402	X5R	4	
Resistor	R3-8		22Ω		0201		6	5%
	R12		100KΩ		0201		1	5%
	R10,R11		0Ω		0201		2	5%
DIODE	D1	RB521S-30			0201		1	
FLASH	F1	GD25LQ80B					1	
CONNECTOR	CNT1		'2458570482018929 +				1	

9-2-3.Main FPC Placement

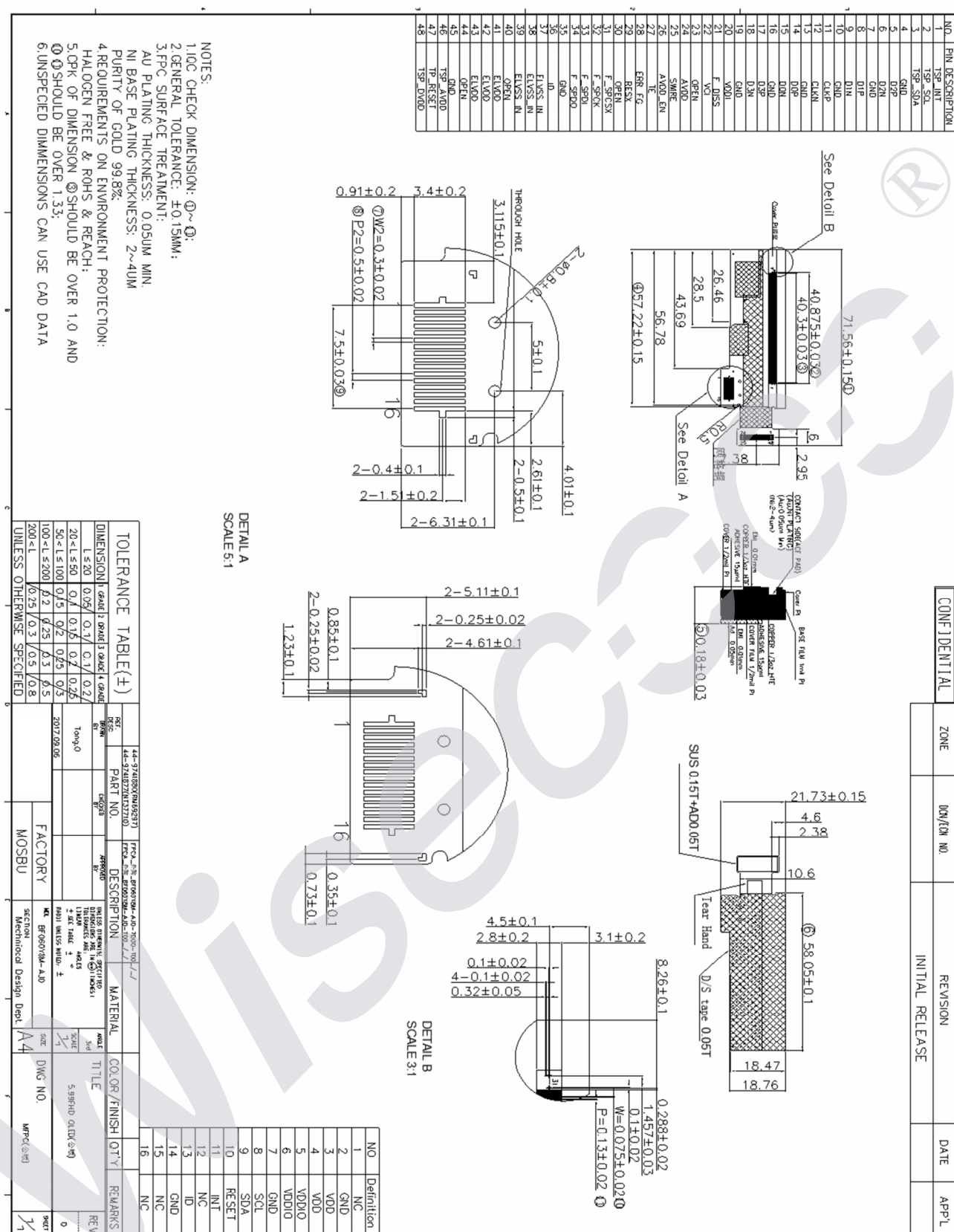


Figure 20

9-3. TSP FPCB Drawing

9-3-1.TSP FPC Schematic Diagram

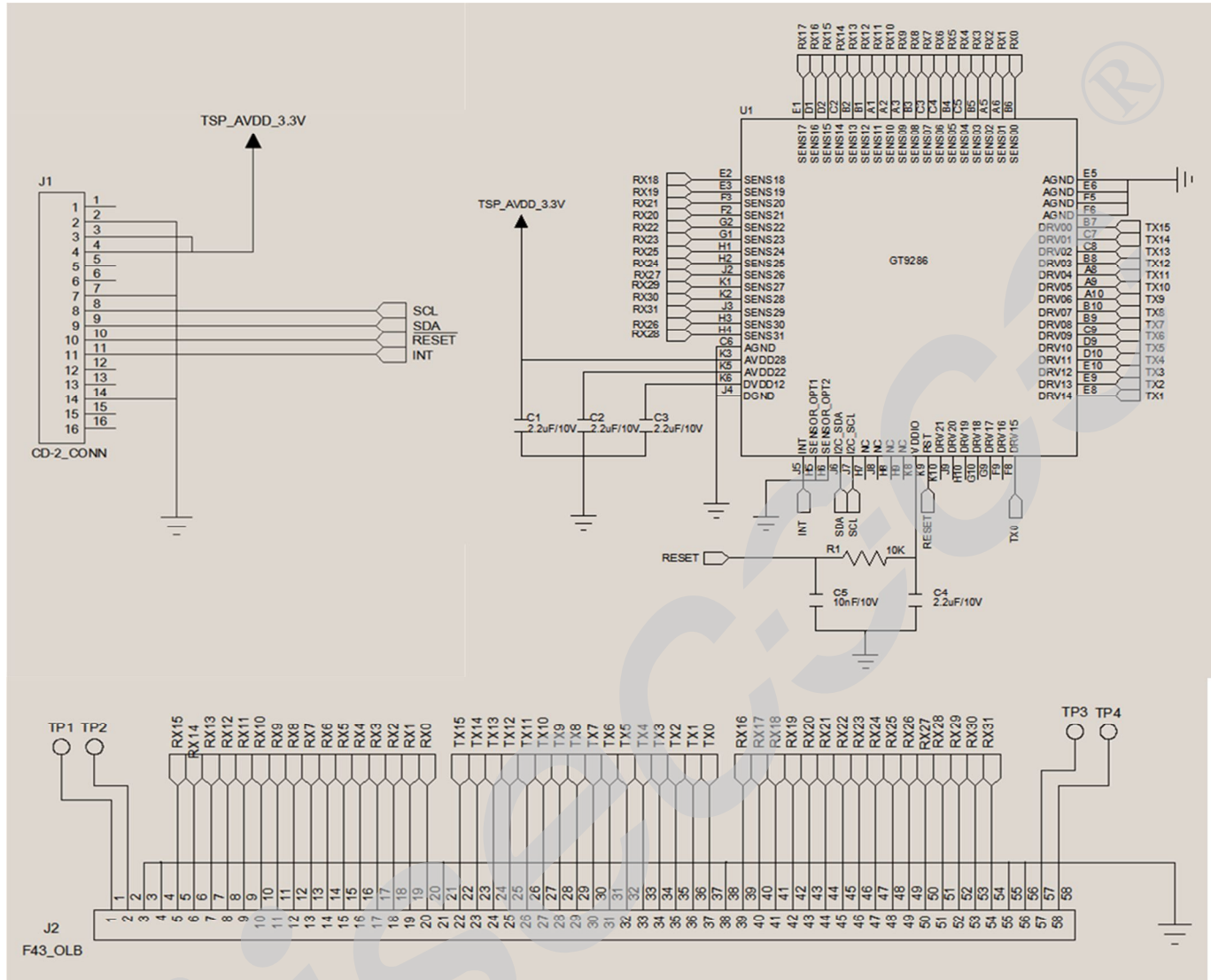


Figure 21

9-3-2.TSP FPC Electronic Part List

Table 15

CATEGORY	REFERENCE	SPECIFICATION	VALUE	耐压值	封装	材质	Q'ty	MAKE
CAPACITOR	C1		2.2uF	10V	0402	X5R	1	
	C2		2.2uF	10V	0402	X5R	1	
	C3		2.2uF	10V	0402	X5R	1	
	C4		2.2uF	10V	0402	X5R	1	
	C5		10nF	10V	0402	X5R	1	
Resistance	R1		10K				1	
U1	U1	GT9286-BGA 5.0*5.0*0.6mm					1	

9-3.3.TSP FPC Placement

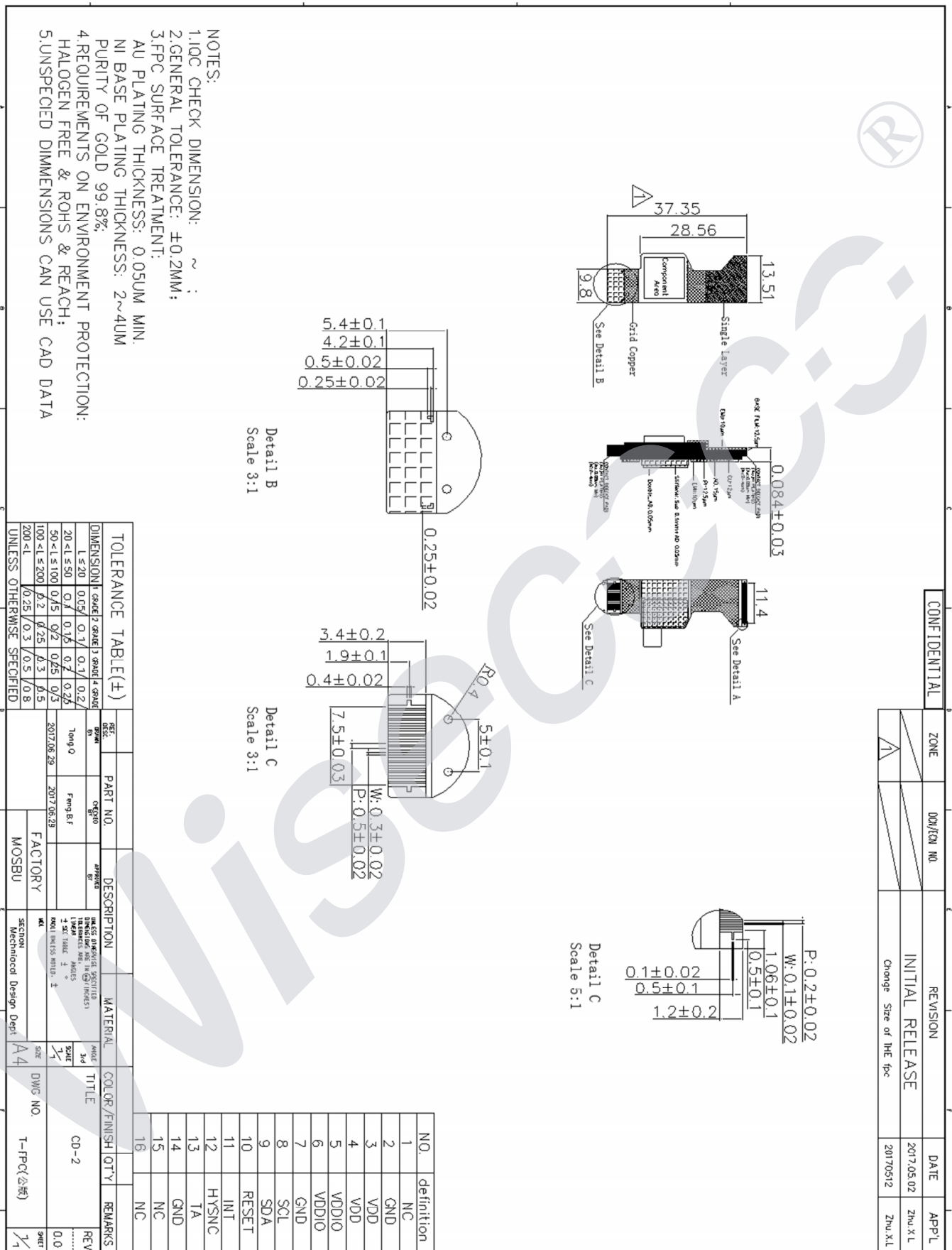


Figure 22

10. Reliability TEST

Table 16

测试项目		测试条件	判定条件
环境 信赖 性	THO	60°C , 90% RH , Operation , 120h	
	LTO	-20°C , Operation , 120h	
	HTO	70°C , Operation , 120h	
	HTS	80°C , Shortage , 120h	
	LTS	-40°C , Shortage , 120h	
	TST	-40°C~85°C , 1h/Cycle , 100Cycle	
电学 信赖 性	非上电模式 (100pF+1500ohm)	Contact : ±6.0kV Air : ±8.0kV 产品显示面向上	
	上电模式 (150pF+330ohm)	Contact : ±4.0kV Air : ±6.0kV 产品显示面向上	
机械 信赖 性	VIB	5-200Hz , 0.015psd , 200-500 , -6db/octave , X/Y/Z each axis 2hrs , total 6hrs	
	Drop	按照产品重量确定高度	

11. Handling Precautions

11-1. Mounting Method

The AMOLED panel module can easily get damaged. Since the module is constructed as to be fixed by utilizing fitting holes in the printed circuit board. Extreme care should be used when handling the AMOLED modules.

11-2. Caution of AMOLED Handling and Cleaning

When cleaning the display surface, use soft cloth solvent as recommended below and wipe gently.

- Ethyl alcohol

Do not wipe the display surface with dry or hard materials that will damage the polarizer surface. Do not use the following solvent.

- Water
- Ketone
- Aromatics

Do not wipe ITO pad area with the dry or hard materials that will damage the ITO patterns. Do not use the following solvent on the pad and prevent it from being contaminated.

- Soldering flux
- Chlorine(Cl), Sulfur(S)
- Spittle, Fingerprint

11-3. Caution against Static Charge

For AMOLED module, use C-MOS LSI drivers, therefore we recommend that you ; Connect any unused input terminal to Panel, do not input any signals before power is turned on, and ground your body, work/assembly areas, assembly equipment to protect against static electricity. It could occur static electricity when taping off the film which protects AMOLED. Against static charge, you should make sure that the product is safe or not by experiment in advance.

11-4. Packing

- The packing principle is that AMOLED module should keep its packing condition at the time of delivery.
- For safety & avoiding the module damage, Carton box must stack the below boxes. When storing the AMOLED after unpacking, note the followings.
- AMOLED module is consisted of flexible materials. It should avoid pressure, especially the back and edge of the product. It should avoid strong impact, and being dropped from a height.
- To prevent modules from degradation, do not operate or store them in a place where they are directly exposed to sunlight or high temperature/humidity.

11-5. Caution for Operation

- If you do not follow normal POWER ON, OFF sequence or abnormal operating, then AMOLED module can be damaged electro-optically and does not recover.
- Response time may extremely delay at a temperature lower than operating range; AMOLED does not normally operate at a high temperature. But this may recover at a proper temperature.
- When you set optimal operating voltage to AMOLED module, you can see the optimal contrast of AMOLED. So, add voltage controllable function at SET Module.
- AMOLED module may not display normally when abnormal power or pressing power is added. Therefore you should secure AMOLED module maximum power at set not to have any pressure affect AMOLED module.
- Electro-chemical reaction may occur when there is humidity on pad; therefore, you should use AMOLED Module below maximum operating humidity.
- AMOLED Module Power Vdd should be designed to protect surge current at SET Module.
- You should not damage connector and cable for AMOLED module assembly by force folding or by applying extreme power.
- AMOLED may not display normally when it is interfered by surrounding elements, therefore you should consider setting design not to damage AMOLED module by surrounding elements.
- To satisfy EMI standards, you should plan your design after considering emitting energy.
- We cannot guarantee display characteristics outside viewing area, therefore your set window should be fixed into viewing area.
- Image-sticking may occur if AMOLED displays same image for a long time, so you need to make a change for AMOLED.
- When remove the window protective film, necessarily need to apply as a way to prevent Cushion and conductive tape Delamination.
 - As an upper Figure, the handler takes off the direction of the arrow to remove the protective film.

11-6. Storage

- Place in a dark place where neither exposure to direct sunlight nor any fluorescent light is permitted and keep at room temperature & room humidity.
- Store with no contact with product surface.
[It is recommended to store them as they have been contained in the inner container when we delivered them.]

11-7. Safety Precautions

- Disassembly or modification may cause electric shock, damages to sensitive part inside of the AMOLED module, dust adhesion, or scratches on the display part.
- In the event that the contents of AMOLED module are on skin, wipe them with a paper towel or gauge and wash the part well, and receive medical attention if necessary.
- Do not use the AMOLED module for the special purpose besides display units.
- Be careful of the glass chips that may cause injury to fingers of skin, when the display part is broken.
- For keeping safe quality from outer exposure or contamination, modules should be consumed within

2 months after unpacking.

11-8. Precautions before us

- In case of any questions about contents of this Specification.
- In case of occurring new problems not mentioned at this Specification.
- In case of your request about income inspection specification change.
- In case of occurring new problem at your driving test.

12. Packing Specification

12-1. Box Pack

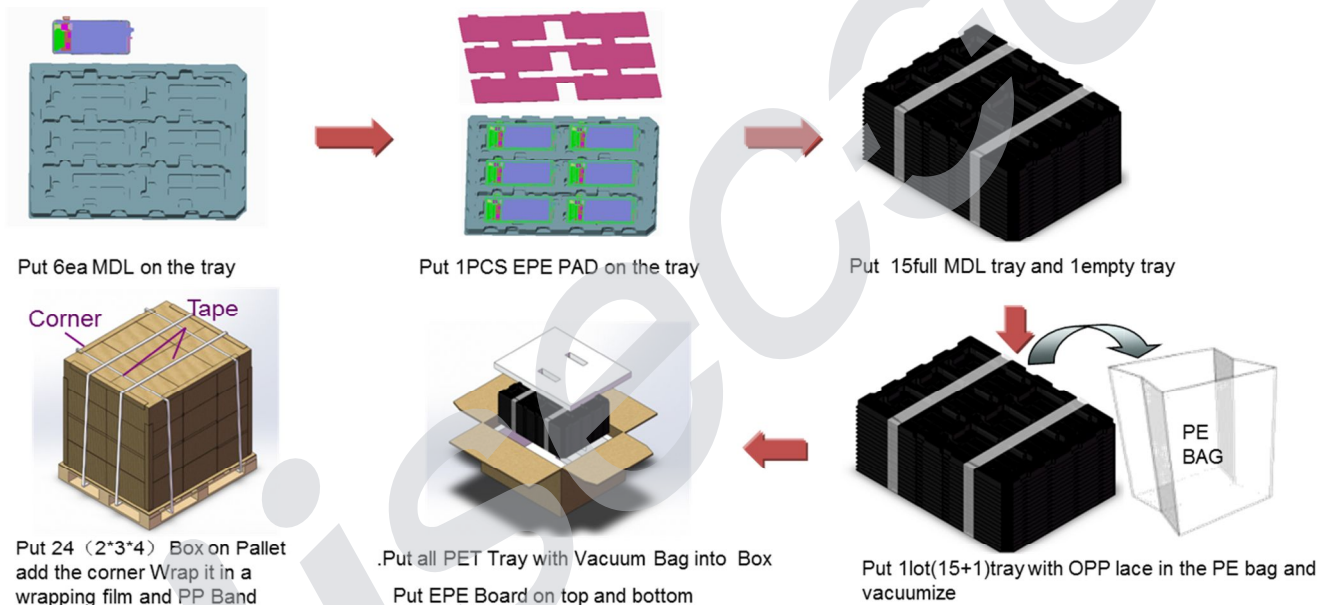


Figure 23